Limits & Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x\to 2}(3x^2-4x+1)$.

$3(4)-8+1=12-8+1=5.$

2 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x\to -1}(x^3+2x-5)$.

$(-1)^3+2(-1)-5=-1-2-5=-8.$

3 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x\to 0}\frac{5x+7}{x+3}$.

$\frac{5(0)+7}{0+3}=\frac{7}{3}.$

4 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x\to 3}\sqrt{x+13}$.

$\sqrt{3+13}=\sqrt{16}=4.$

5 **[Foundational]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x\to 3}\frac{x^2-9}{x-3}$.

$\frac{(x-3)(x+3)}{x-3}=x+3\to 6.$

6 **[Foundational]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x\to -2}\frac{x^2+5x+6}{x+2}$.

$\frac{(x+2)(x+3)}{x+2}=x+3\to 1.$

7 **[Foundational]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x\to 1}\frac{x^2-1}{x^2-3x+2}$.

$\frac{(x-1)(x+1)}{(x-1)(x-2)}=\frac{x+1}{x-2}\to\frac{2}{-1}=-2.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{3x^2+2x}{x^2-5}$.

The degrees match, so the limit is the ratio of leading coefficients: 3.

9 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{2x+7}{x^2+1}$.

The denominator has higher degree, so the limit is 0.

10 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{5x^3-2}{x^3+4x}$.

The degrees match, so the limit is the ratio of leading coefficients: 5.

11 [Intermediate] • Target: One-sided limits

A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x\to 2^-} f(x)$.

Using the first rule: $\lim_{x\to 2^-} f(x) = 2 + 1 = 3$.

12 [Intermediate] • Target: One-sided limits

Using the same piecewise function, find $\lim_{x\to 2^+} f(x)$.

Using the second rule: $\lim_{x\to 2^+} f(x) = 2(2) - 1 = 3$.

13 [Intermediate] • Target: One-sided limits

Using the same piecewise function, does $\lim_{x\to 2} f(x)$ exist? State its value.

Yes — both one-sided limits equal 3, so $\lim_{x\to 2} f(x) = 3$.

14 [Intermediate] • Target: The power rule

Differentiate $f(x) = x^5$.

$f'(x) = 5x^4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The power rule • Cross-Lesson: 4.01 Trigonometry

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

$f'(x) = 12x^3 - 4x$.

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16 [Challenging] • Target: The power rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Differentiate $f(x) = \sqrt{x}$.

$f(x) = x^{1/2}$, so $f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.

17 [Challenging] • Target: The power rule • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = \frac{1}{x^2}$.

$f(x) = x^{-2}$, so $f'(x) = -2x^{-3} = -\frac{2}{x^3}$.

18 [Challenging] • Target: The product rule • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = x^2(3x + 1)$ using the product rule.

$f'(x) = (2x)(3x + 1) + x^2(3) = 6x^2 + 2x + 3x^2 = 9x^2 + 2x$.

19 [Challenging] • Target: The product rule • Cross-Lesson: 9.01 Statistics & Probability

Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.

$f'(x) = (1)(x - 5) + (x + 2)(1) = x - 5 + x + 2 = 2x - 3$.

20 [Challenging] • Target: The product rule • Cross-Lesson: 1.01 Algebra & Equations

Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.

$f'(x) = (2)(x^2 + 3) + (2x - 1)(2x) = 2x^2 + 6 + 4x^2 - 2x = 6x^2 - 2x + 6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.

a. $y = 3x - 9$

b. $y = 4x - 3$

c. $y = 4x + 9$

d. $y = 4x - 9$ ✓

$f(3) = 3$, $f'(3) = 4$: $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.

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22 [Foundational] • Target: Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.

a. 4 ✓

b. 2

c. 1

d. 8

The tangent line rises 4 for every 1 across.

23 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to -2} (x^3 - 3x + 4)$.

a. 10

b. -2

c. 2 ✓

d. -10

$(-2)^3 - 3(-2) + 4 = -8 + 6 + 4 = 2$.

24 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$ ✓

$\frac{16 - 12}{9} = \frac{4}{9}$.

25 [Foundational] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x \to -1} \frac{x^2 - 1}{x^2 + 3x + 2}$.

a. -2 ✓

b. 2

c. -1

d. $-\frac{1}{2}$

$\frac{(x-1)(x+1)}{(x+1)(x+2)} \to \frac{x-1}{x+2} = \frac{-2}{1} = -2$.

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26 [Foundational] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{4x^2+3x}{2x^2-7}$.

a. 2 ✓

b. 0

c. $\frac{3}{7}$

d. 4

Same degree: ratio of leading coefficients $= \frac{4}{2} = 2$.

27 [Foundational] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{2x^2+1}{5x^3-4}$.

a. ∞

b. $\frac{2}{5}$

c. 0 ✓

d. 5

The denominator's degree is higher: the limit is 0.

28 [Foundational] • Target: One-sided limits

A function is defined by $f(x) = 2x$ for $x \le 1$, and $f(x) = x + 3$ for $x > 1$. Find $\lim_{x\to 1^-} f(x)$.

a. 1

b. 2 ✓

c. 3

d. 4

Using $x \le 1$: $f(x) \to 2(1) = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: One-sided limits

Using the same piecewise function, find $\lim_{x\to 1^+} f(x)$, and state whether $\lim_{x\to 1} f(x)$ exists.

a. 4**; does not exist** ✓

b. 4; exists

c. 2; exists

d. 2; does not exist

Right limit $= 1 + 3 = 4$. Since the left limit (2) differs from the right limit (4), the two-sided limit does not exist.

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30 [Intermediate] • Target: The power rule

Differentiate $f(x) = 4x^5 - 3x^3 + 2x$.

a. $4x^4 - 3x^2 + 2$

b. $20x^4 - 3x^2 + 2$

c. $20x^4 - 9x^2$

d. $20x^4 - 9x^2 + 2$ ✓

Differentiate each term: $20x^4 - 9x^2 + 2$.

31 [Intermediate] • Target: The power rule

Differentiate $f(x) = 6x^2 - 4x + 9$.

a. $6x - 4$

b. $12x + 9$

c. $12x - 4x$

d. $12x - 4$ ✓

Differentiate each term: $12x - 4$.

32 [Intermediate] • Target: The product rule

Differentiate $f(x) = (x^2 + 1)(x^2 - 1)$ using the product rule.

a. $4x^3$ ✓

b. $4x$

c. $2x^3$

d. $4x^3 - 2x$

$f'(x) = 2x(x^2 - 1) + (x^2 + 1)(2x) = 2x^3 - 2x + 2x^3 + 2x = 4x^3$.

33 [Intermediate] • Target: The product rule

Differentiate $f(x) = x^2(x^3 - 4)$ using the product rule.

a. $5x^4 - 4$

b. $3x^4 - 8x$

c. $5x^4 - 8x$ ✓

d. $5x^4 - 8$

$f'(x) = 2x(x^3 - 4) + x^2(3x^2) = 2x^4 - 8x + 3x^4 = 5x^4 - 8x$.

34 [Intermediate] • Target: The chain rule

Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$ ✓

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

$f'(x) = 2(x^3 + 2) \cdot 3x^2 = 6x^2(x^3 + 2)$.

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35 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = \cos x - \sin x$.

a. $\sin x + \cos x$

b. $-\sin x - \cos x$ ✓

c. $\sin x - \cos x$

d. $-\sin x + \cos x$

$\frac{d}{dx}[\cos x] = -\sin x$, $\frac{d}{dx}[-\sin x] = -\cos x$.

36 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = 5\cos x$.

a. $5\sin x$

b. $-5\cos x$

c. $5\cos x$

d. $-5\sin x$ ✓

$\frac{d}{dx}[5\cos x] = -5\sin x$.

37 [Intermediate] • Target: Derivatives of trigonometric and exponential functions

Differentiate $f(x) = e^x - x^3$.

a. $e^x - 3x$

b. $e^x - x^2$

c. $xe^{x-1} - 3x^2$

d. $e^x - 3x^2$ ✓

$\frac{d}{dx}[e^x] = e^x$, $\frac{d}{dx}[-x^3] = -3x^2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Derivatives of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = 2e^x + \sin x$.

a. $2e^x + \cos x$ ✓

b. $2e^x + \sin x$

c. $e^x + \cos x$

d. $2e^x - \cos x$

$\frac{d}{dx}[2e^x] = 2e^x$, $\frac{d}{dx}[\sin x] = \cos x$.

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39 [Challenging] • Target: Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

Find the second derivative of $f(x) = x^4 + 2x^2$.

a. $12x^2$

b. $4x^3 + 4x$

c. $12x^2 + 4$ ✓

d. $12x^2 + 4x$

$f'(x) = 4x^3 + 4x$; $f''(x) = 12x^2 + 4$.

40 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.

a. 10 ✓

b. -10

c. 30

d. 20

$h'(t) = -10t + 30$; $h'(2) = 10$ m/s.

41 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 2.01 Functions & Transformations

A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.

a. 8 dollars

b. 4 dollars

c. 20 dollars

d. 12 **dollars** ✓

$C'(x) = 0.04x + 8$; $C'(100) = 12$.

42 [Challenging] • Target: Word problems: rates of change • Cross-Lesson: 3.01 Sequences & Series

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$ ✓

b. $t = 5$

c. $t = 6$

d. $t = 2$

$h'(t) = -10t + 30 = 0 \Rightarrow t = 3$ seconds.

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43 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 4.01 Trigonometry

In a laboratory calibration sequence, evaluate $\lim_{x\to -5}(-3x^2-5x+4)$.

a. -49

b. -43

c. 46

d. -46 ✓

$-3(-5)^2+25+4=-46$.

44 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\lim_{x\to -4}(x^2-x-2)$.

a. 18 ✓

b. 21

c. -18

d. 15

$(-4)^2+4-2=18$.

45 [Challenging] • Target: Limits & Derivatives • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\lim_{x\to 0}(-2x^2+x-2)$.

a. 1

b. -2 ✓

c. 2

d. -5

$-2(0)^2-2=-2$.

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